

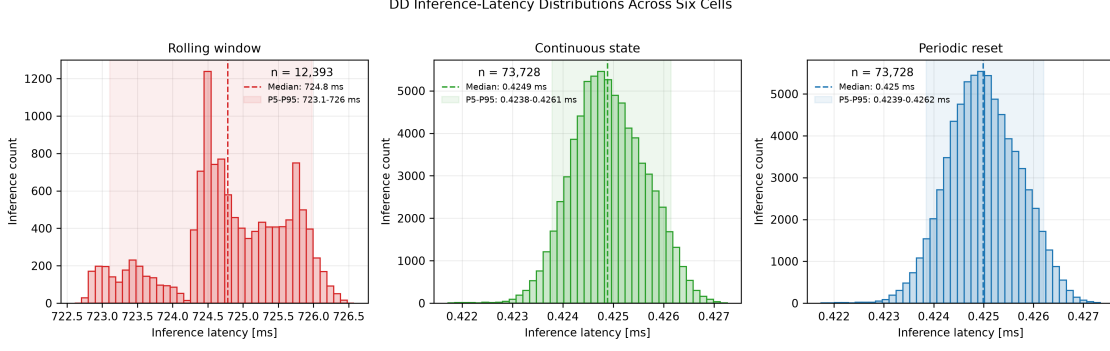


Figure A.2: DD latency distributions over the six hardware cell sequences. Exact rolling-window execution is shown separately from continuous-state and periodically reset execution because their scales and recurrent semantics differ substantially.

A.1.1 HECM lookup-table sensitivity

The cross-model benchmark evaluates one fixed HECM implementation whose lookup surfaces are identified exclusively from training and validation data. The supplementary HECM-only analysis tests whether its disturbance response depends strongly on local lookup calibration. The R_i , R_1 , and R_2 surfaces are jointly scaled by -10% or $+10\%$. The τ_1 and τ_2 surfaces are likewise scaled by -10% or $+10\%$, while the OCV surfaces are shifted by -10 mV or $+10$ mV. These six one-at-a-time perturbations and the nominal lookup are evaluated across 22 measurement and signal-integrity subcases and the paired initialization intervention. All original seeds and 16 frozen windows are retained. The complete design comprises 9184 HECM executions.

Main-campaign disturbances use their lookup-matched baseline, the one-hour dropout uses a lookup-matched 48-h baseline, and initialization recovery uses a separate paired baseline. For every disturbance, the lookup interaction is the scenario Δ MAE under the perturbed lookup minus the corresponding Δ MAE under the nominal lookup. Results are summarized within cells before the same equal-weight cell bootstrap used by the primary benchmark is applied.

Local lookup changes shift baseline HECM MAE between 0.0314 and 0.0373, compared with 0.0337 for the nominal lookup. Across the 22 disturbance subcases, the largest absolute interaction is 0.006116 MAE. It occurs for the -0.05 A current offset under $+10\%$ time-constant scaling, and its cell-bootstrap interval includes zero. This interaction is small relative to the nominal adverse current-offset penalty